

KAMALESH SANKARANARAYANAN

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🌐 LinkedIn 🐙 GitHub 🌐 Portfolio

ABOUT ME

Final-year Computer Science and Engineering student with hands-on experience in software development, AI, and IoT. Skilled in C++, Python, Flask, MySQL, Data Structures & Algorithms, LLMs, and cloud technologies. Seeking Software Engineer or Backend Engineer opportunities.

EDUCATION

PSG Institute of Technology and Applied Research

CGPA: 8.61

B.E. Computer Science and Engineering

Saraswathi Matric Hr. Sec. School

97.3%

Higher Secondary Certificate (HSC)

SKILLS

Languages: C, C++, Java, Python, JavaScript, TypeScript

Frameworks: Spring Boot, React, Flask, FastAPI, LangGraph

AI/ML: LLMs, Hugging Face, Whisper, OpenRouter, Prompt Engineering

Databases: PostgreSQL, MySQL, SQLite

Cloud: Oracle Cloud Infrastructure (OCI), Firebase, Render

Tools: Git, GitHub, Docker, FFmpeg, HLS, VS Code

KEY PROJECTS

LLM Guided Code Optimization Tool

- Built an LLM-based tool for code optimization and refactoring in C programs.
- Reduced debugging effort using real-time AI-assisted suggestions.

IntelliStream – Intelligent Adaptive Video Streaming Research Platform

- Built a full-stack adaptive video streaming platform using Java, Spring Boot, React, PostgreSQL, FFmpeg, and HLS with asynchronous multi-resolution video transcoding.
- Implemented intelligent ABR algorithms, JWT authentication, role-based access, authenticated media delivery, watch history, and experiment benchmarking under varying network conditions.

IoT Based Water Quality Monitoring System

- Built a real-time system to monitor pH, turbidity, temperature, and TDS using Arduino and sensors.
- Developed a dashboard with alerts for unsafe water conditions.

Laptop Safety & User Health Monitoring System

- Developed a monitoring system for temperature, lighting, screen usage, and user presence.
- Generated alerts for overheating, low light, and prolonged screen exposure.

INTERNSHIPS

Software Intern – RND Optimisers

June 2026

- Designed and developed an AI Meeting-to-Execution Agent using LangGraph, Flask, Whisper, and OpenRouter.
- Built an end-to-end workflow for transcription, task extraction, progress tracking, and automated email reminders.

Python Developer Intern – Gustavovalley Technovations

- Worked on real-world programming tasks, debugging, and Python-based development.

Data Science Intern – Elewayte

- Performed data analysis and visualization on real datasets and implemented basic machine learning models.

CERTIFICATIONS & COURSES

- Oracle Cloud Infrastructure 2025 Certified Foundations Associate
- Design and Analysis of Algorithms – NPTEL
- Machine Learning A-Z: AI, Python – Udemy
- Application Development using Arduino – JRM Technologies

ACHIEVEMENTS & ACTIVITIES

- ITech Ideathon 2026 Finalist.
- Advanced to the Interview Stage of HackWithInfy 2026 after clearing the coding rounds conducted by Infosys.
- NSS Volunteer – Participated in community service and awareness programs.
- **Areas of Interest:** Data Structures & Algorithms, OS, DBMS, Computer Networks, OOPS, Machine Learning, IoT.